

DIVYAPRADA G

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EDUCATION

Siddaganga Institute of Technology, Karnataka

B.E. in Artificial Intelligence & Data Science

Expected Graduation: Jun 2027

Coursework: DSA, OOP, Operating Systems, DBMS, Computer Networks

HMS Polytechnic, Tumkur

Diploma in Computer Science & Engineering

Graduated: 2024

CGPA: 9.77 / 10 | Distinction

TECHNICAL SKILLS

Languages: Java (Multithreading, Streams, Concurrency), Python, C++, SQL

Backend & APIs: Java, REST APIs, Microservices, Concurrency, Rate Limiting, API Design

Distributed Systems: Redis, Apache Kafka, Real-Time Stream Processing, Consistent Hashing, Systems Architecture

AI Infrastructure: LLM Integrations, Gemini API, Claude API, LangChain, OpenAI SDK, Vertex AI, Custom RAG Systems

Databases: MySQL (Indexing, Query Optimization), MongoDB, BigQuery

Architecture & Concepts: LLD/HLD, SOLID Principles, Cache Locality, Scalability, CAP Theorem, System Design

Tools & DevOps: Docker, Kubernetes, Git, GitHub Actions, Linux, Postman, CI/CD

PROJECTS

Custom Distributed Time-Series Database (TSDB) | *Java, Netty, Java NIO*

[GitHub](#)

- Built an industrial-grade columnar TSDB engine from scratch using Java NIO to ingest and query millions of metrics per second.
- Implemented Facebook's Gorilla Compression algorithm, minimizing JVM heap footprint by **10x** and eliminating GC spikes; built a Vectorized Execution Engine optimizing cache locality to improve aggregation throughput by **40%** over row-based models.
- Designed asynchronous write buffering and immutable storage components for efficient persistence and reduced contention during concurrent writes.

Distributed LLM Inference & Custom Vector Search Engine | *Python, C++, gRPC, SIMD*

[GitHub](#)

- Engineered a high-throughput HNSW graph vector search engine from scratch over datasets scaling past **1M+ vectors**; used Product Quantization to compress 768-dimensional embeddings by **4x** with SIMD hardware-level cosine similarity calculations at **sub-10ms** latency.
- Built a concurrent Dynamic Batching Coordinator and PagedAttention KV-Cache manager via gRPC nodes, eliminating memory fragmentation and boosting token throughput metrics by **2.5x**.
- Implemented Speculative Decoding pipelines pairing a lightweight draft model alongside a target LLM, shaving inference latency by **45%**.

Adaptive Real-Time Traffic System | *Java, Kafka, Event-Driven Architecture*

[GitHub](#)

- Designed a distributed event-driven system handling real-time streaming telemetry data from **10K+ simulated vehicles** through Apache Kafka pipelines with isolated dead-letter queues, validating **99.9%** delivery stability.
- Reduced bottleneck detection latency by **60%** against baseline polling; verified fault tolerance via 3-node cluster failures showing an automatic recovery sequence under **<2s**.

EXPERIENCE

Software Engineer Intern

Jan 2024 – Apr 2024

PraLoTech Solutions LLP

- Optimized backend response times by **30%** for a Java enterprise application by constructing targeted composite indexes and restructuring legacy N+1 data queries into high-performance batched join operations.
- Owned end-to-end design and deployment for 3 core functional features including seat reservation loops, multi-tenant booking state, and profile management under Git CI workflows with zero system regressions.

Software Engineering Instructor

2023 – 2024

HMS Polytechnic, Tumkur

- Delivered structured coursework on system design principles, SDLC models, and debugging strategies to **60+ engineers**; mentored development pipelines for 5 production-style applications, elevating code quality marks by **25%**.

ACHIEVEMENTS

- Google Student Ambassador (GSA 2026)** | *Google Developer Ecosystem*: Selected to represent the student developer community and support initiatives involving Google technologies, Gemini, cloud, developer learning, and open-source.
- Campus Lead & Open-Source Contributor** | *Open Source Connect India 2026*: Representing Siddaganga Institute of Technology and supporting open-source collaboration, technical learning, mentorship, developer connections, and contribution opportunities.
- Google Cloud Arcade Facilitator** | *Technical Training*: Led technical upskilling workshops for cloud architectures, focusing on generative AI and serverless technologies.
- LeetCode Competitive Programming Rank: Global 700+**: Solved over **2,626+ algorithmic problems** including **600+ Hard** and **1,370+ Medium**, covering trees, graphs, dynamic programming, and system-oriented problem solving.
- Top 10 — Smart India Internal Hackathon 2025** (*Out of 165+ Engineering Teams*): Architected a scalable real-time emergency disaster routing engine during an intensive 48-hour development sprint.